

Beyond Instance-Level Self-Supervision in 3D Multi-Modal Medical Imaging

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Abstract

Self-supervised pre-training methods in medical imaging typically treat each individual as an isolated instance, learning representations through augmentation-based objectives or masked reconstruction. They often do not adequately capitalize on a key characteristic of physiological features: anatomical structures maintain consistent spatial relationships across individuals (instances), such as the thalamus being medial to the basal ganglia, regardless of variations in brain size, shape, or pathology. We propose leveraging this cross-instance topological consistency as a supervisory signal. The challenge arises from the inherent variability in medical imaging, which can differ significantly across instances and modalities. To tackle this, we focus on two alignment regimes. (i) Intra-instance: with pixel-level correspondences available, a cross-modal triplet objective explicitly preserves local neighborhood topology. (ii) Inter-instance: without such supervision, we derive pseudo-correspondences to control partial neighborhood alignment and prevent topology collapse across modalities. We validate our approach across 7 downstream multi-modal tasks, achieving average improvements of 1.1% and 5.94% in segmentation and classification tasks, respectively, and demonstrating significantly better robustness when modalities are missing at test time.

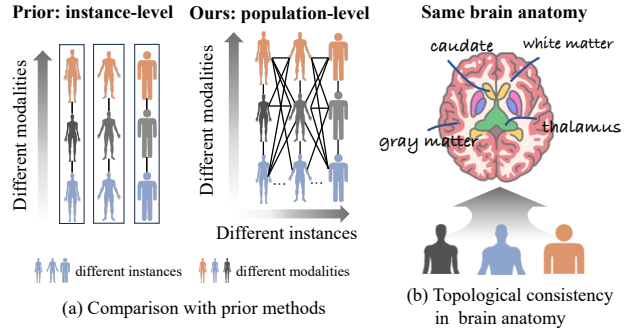


Figure 1. (a) Comparison with prior methods. Prior work learns instance-level representations via self-supervision, treating each individual independently. In contrast, our method learns cross-individual, multi-modal representations that capture relations across individuals. (b) Topological consistency. Human anatomy shares population-level topology (common structures and spatial layout), offering a natural prior for representation learning.

1. Introduction

Self-supervised learning (SSL) has become the dominant paradigm for visual representation learning, with contrastive learning (He et al., 2020), knowledge distillation (Grill et al., 2020), and masked autoencoding (He et al., 2022) achieving strong results across domains. In medical imaging, approaches are adapted with domain priors such as anatomical invariance (Pan et al., 2025; Jiang et al., 2023) and cross-modal consistency (Rui et al., 2025; Tan et al., 2025). Yet a common limitation remains: nearly all methods treat each individual as an independent instance, learning by contrasting augmented views or reconstructing masked regions within the same individual. While effective for ensuring robustness, those paradigms overlook a key characteristic of physiological features: human anatomy exhibits a stable, shared structure across individuals (instances). As a result, it underutilizes cross-instance regularities that could further enhance the quality of the learned representations.

Building on this, Fig. 1 (b) highlights the topological consistency observed in brain anatomy across different individuals. Despite considerable variations in brain size, shape, and pathology, the spatial relationships between anatomical structures remain remarkably stable. For example, the thalamus consistently lies medially to the basal ganglia, ventricles are adjacent to deep gray matter structures, and

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cortical regions maintain their relative neighborhood patterns. This consistency, crucially, exists not at the pixel level but in the structural relationships between regions (Segobin et al., 2024). Previous work (He et al., 2023) leverages explicit geometric consistency to align inter-image representations in self-supervised learning, typically using known or learned transformation matrices (e.g., affine transforms). However, implicit transformations or discrepancies, such as those arising across modalities, are difficult to parameterize or recover with an explicit matrix, making such formulations less applicable in cross-modal or cross-instance scenarios.

To address this, we introduce TACO, a pre-training framework that leverages **T**opology-Aware **C**onsistency across individuals as a form of multi-modal and multi-instance supervision. The key idea in this paper is to ensure that if two patches belong to the same region in scan *A*, their corresponding features from scan *B* should also co-cluster as the same region in *B*’s representation space. This form of supervision goes beyond traditional instance-level learning: instead of focusing solely on the appearance of individual structures, the model learns the relative positioning of structures within their broader anatomical context, as shown in Fig. 1(a). To achieve this, we use a dual alignment approach: intra-instance and inter-instance consistency. For intra-instance alignment, we apply a cross-modal triplet objective, preserving local neighborhood topology within the same individual across modalities. For inter-instance alignment, we use unsupervised Mutual Nearest Neighbor (MNN) search to establish pseudo-correspondences, ensuring partial neighborhood alignment across instances.

We conduct extensive experiments across 7 tasks, demonstrating that our method achieves state-of-the-art (SOTA) performance and exhibits significantly better robustness in missing-modality scenarios, where some imaging sequences are unavailable at test time. To further validate our approach, we present t-SNE visualizations of the learned embedding space. The results show that our method effectively clusters feature tokens by anatomical identity across different individuals and modalities, which aligns well with our intended goal. In contrast, baseline self-supervised methods tend to cluster features by individual identity rather than anatomical correspondence, indicating that they capture instance-specific patterns rather than generalizable anatomical structures. Our main contributions are as follows:

- (1) To the best of our knowledge, we are the first to jointly model inter-instance and inter-modality topological consistency as a supervisory signal for medical image pre-training;
- (2) We develop TACO, a pre-training framework that integrates intra-instance objectives (cross-modal alignment) with inter-instance constraints (cross-instance topological alignment) through partial correspondence matching;

- (3) We demonstrate consistent and significant improvements over SOTA medical SSL baselines across 7 downstream multi-modal tasks, including those with challenging missing-modality scenarios. Additionally, visualizations of region features provide compelling evidence of the effectiveness of our proposed method, further validating its potential for enhancing medical image pre-training. Code will be publicly available at <https://github.com/Ashespt/TACO>.

2. Related Work

3D medical self-supervised learning. Most 3D SSL work targets CT and MRI. Early approaches (Wang et al., 2023c; Hatamizadeh et al., 2021; Wald et al., 2025) follow reconstruction- or contrastive-based objectives, mirroring general-purpose SSL. More recent methods incorporate anatomical invariance to enforce geometric consistency (He et al., 2023; Pan et al., 2025). While these techniques deliver strong results, they typically ignore multi-modality settings and largely confine supervision to intra-instance (within-scan) structure.

Pre-training in 3D multi-modal medical imaging. Medical pretraining has emerged as a promising route to cross-modality generalization (Chen et al., 2020; Tang et al., 2022; Jiang et al., 2023; Wu et al., 2024; Tan et al., 2025). For example, (Jiang et al., 2023) captures class-level anatomical invariants but overlooks patient-specific cues; Brain-MVP (Rui et al., 2025) distills modality-aware templates; and PUIR (Tan et al., 2025) emphasizes personal-level information while maintaining cross-modal alignment. However, these approaches still tend to treat scans in isolation, limiting the use of cross-patient topology and inter-instance relations that are crucial for robust multi-modal representation learning.

Missing modality segmentation. Most current work that focuses on generalization across modalities introduces tasks such as modality transfer and missing modality segmentation using structural modalities (Flair, T1, T2, and T1ce) of MRI for brain tumor segmentation (Zhao et al., 2022). There are three main types of approaches to missing modality segmentation. Knowledge distillation-based approaches transfer knowledge from models with complete modality information (teachers) to models with missing modality information (students) (Chen et al., 2021; Wang et al., 2023b). mmFormer (Zhang et al., 2022) and RFNet (Ding et al., 2021) recover missing information by leveraging the multi-modal latent feature space. Domain adaptation-based methods aim to reduce the gap between models with complete and incomplete modalities by aligning their domains (Wang et al., 2021).

Topology in medical imaging Topological consistency is a natural and crucial property in medical images (Iglesias

et al., 2023; Fischl, 2012). TopoLoss (Hu et al., 2019) introduces a differentiable, topology-preserving loss and demonstrates its effectiveness for pathology segmentation. Subsequent topology-aware methods primarily target tubular structures (Wen et al., 2025; Elsayed et al., 2025). However, very few works extend topological consistency to the unsupervised/self-supervised setting, despite the fact that human anatomy exhibits an invariant topological structure across patients.

3. Preliminary

Notation. The data set comprises medical images collected from a set of distinct individuals H , including multiple modalities M in each individual $h \in H$. Thus, we denote the dataset as $X = \{x_m^h | h \in H, m \in M\}$, where x_m^h is the image of the individual h in modality m . Given the transformer encoder $\mathcal{E}$, the corresponding output tokens are $Z_m^h = \{z_{m,k}^h\}_{k=1}^K = \mathcal{E}(x_m^h)$, where K is the token number.

For every input x_m^h , we exploit the standard autoencoder pipeline with encoder $\mathcal{E}$ and decoder $\mathcal{D}$ as:

$$\hat{x}_m^h = \mathcal{D}(\mathcal{E}(x_m^h)). \quad (1)$$

where, $\hat{x}_m^h$ is the reconstructed image. The reconstruction objective is

$$\mathcal{L}_{uni} = \|\hat{x}_m^h - x_m^h\|_2^2. \quad (2)$$

Instance-agnostic Multi-modal Neighborhood Ranking Consistency. Biological organs exhibit stable anatomical structure, with consistent spatial relationships observed across modalities and individuals. This consistency motivates us to learn representations that are invariant with respect to modality and instance, while being governed by spatial relationships. To this end, we propose the Instance-agnostic Cross-Modal Neighborhood Ranking Consistency (IM-NRC) assumption for medical multi-modal alignment. Specifically, given two modalities, i and j , corresponding to individuals h and g respectively, we assume the existence of a mapping $C_{i \rightarrow j}^{h \rightarrow g} : Z_i^h \rightarrow Z_j^g$ that establishes the spatial correspondence between Z_i^h and Z_j^g . Then, the model should thus adhere to the following assumption:

Assumption 3.1 (IM-NRC). Given patch token $z_{i,k}^h \in Z_i^h$, if $z_{i,k}^h$ is located in decision region $R_{i,h}^\alpha = \{z_i^h | \sigma_{i,h}(z_i^h) = \alpha, z_i^h \in Z_i^h\}$, then $z_{j,k}^g = C_{i \rightarrow j}^{h \rightarrow g}(z_{i,k}^h)$ should be in region $\{C_{i \rightarrow j}^{h \rightarrow g}(z_i^h) | z_i^h \in R_{i,h}^\alpha\} \subseteq R_{j,g}^\alpha = \{z_j^g | \sigma_{j,g}(z_j^g) = \alpha, z_j^g \in Z_j^g\}$, where $\sigma_{i,h}$ and $\sigma_{j,g}$ are classifiers or discriminators.

In the unsupervised setting, we cannot explicitly obtain $\sigma_{i,h}$ and $\sigma_{j,g}$. Thus, unsupervised methods, such as nearest neighbors, can be leveraged to estimate the decision

regions. Under the IM-NRC assumption, if $\mathcal{P}_{i,h}(k)$ represents the set of indices of neighborhood points for $z_{i,k}^h$, then $\{o | z_{j,o}^g = C_{i \rightarrow j}^{h \rightarrow g}(z_{i,l}^h), l \in \mathcal{P}_{i,h}(k)\}$ should correspond to the set of indices of neighborhood points for $z_{j,k}^g$. The same assumption applies to the non-neighborhood region. This indicates that the optimal distributions of the tokens from Z_i^h and Z_j^g should remain consistent semantics and geometry under the correspondence $C_{i \rightarrow j}^{h \rightarrow g}$. Therefore, to regularize the feature distribution across instances and modalities, we exploit the contrastive metric learning method, such as triplet loss, to constrain the semantic and geometrical consistency, where the triplet in Z_j^g is defined by the geometry in tokens Z_i^h under the correspondence $C_{i \rightarrow j}^{h \rightarrow g}$.

4. Method

This paper aims to learn modality-invariant, population-level representations for medical data that preserve the intrinsic local topology of each individual while ensuring cross-modal topological consistency. To achieve this, we propose intra- and inter-instance cross-modal alignment methods as shown in Fig. 2. In our approach, $\mathcal{L}_{intra}$ and $\mathcal{L}_{inter}$ are employed to regularize cross-modality and cross-instance features in a self-supervised manner by maintaining neighborhood-ranking consistency and enforcing Assumption 3.1. Given the reconstruction objective (2), our overall training objective can be formulated as

$$\mathcal{L}_{total} = \mathcal{L}_{uni} + \mathcal{L}_{intra} + \mathcal{L}_{inter}, \quad (3)$$

where $\mathcal{L}_{intra}$ and $\mathcal{L}_{inter}$ will be introduced in the following part of this section. Specifically, the intra-instance loss $\mathcal{L}_{intra}$ is described in Section 4.1, while the inter-instance loss $\mathcal{L}_{inter}$ is outlined in Section 4.2.

4.1. Intra-instance Constraint

Given data x_i^h and x_j^h from individual h , we have the latent features $Z_i^h = \mathcal{E}(x_i^h)$ and $Z_j^h = \mathcal{E}(x_j^h)$. Following the geometric equivalence assumption (Mao et al., 2022), we assume that correspondences among latent token features mirror the patch-level correspondences in the image domain. Let $C_{i \rightarrow j}^h$ denotes the predefined token-level correspondence that aligns the representations Z_i^h and Z_j^h for individual h . Specifically, as the shapes and spatial locations of structures maintain alignment across modalities for the same individual, we take $C_{i \rightarrow j}^h$ to be the identity on positional indices.

The key insight behind intra-instance cross-modal alignment is that local neighborhoods must be preserved across modalities under the correspondence $C_{i \rightarrow j}^h$. This implies that if tokens are close in Z_i^h , their corresponding tokens should also remain close in Z_j^h , and similarly, if tokens are far apart in Z_i^h , their corresponding tokens should remain distant in

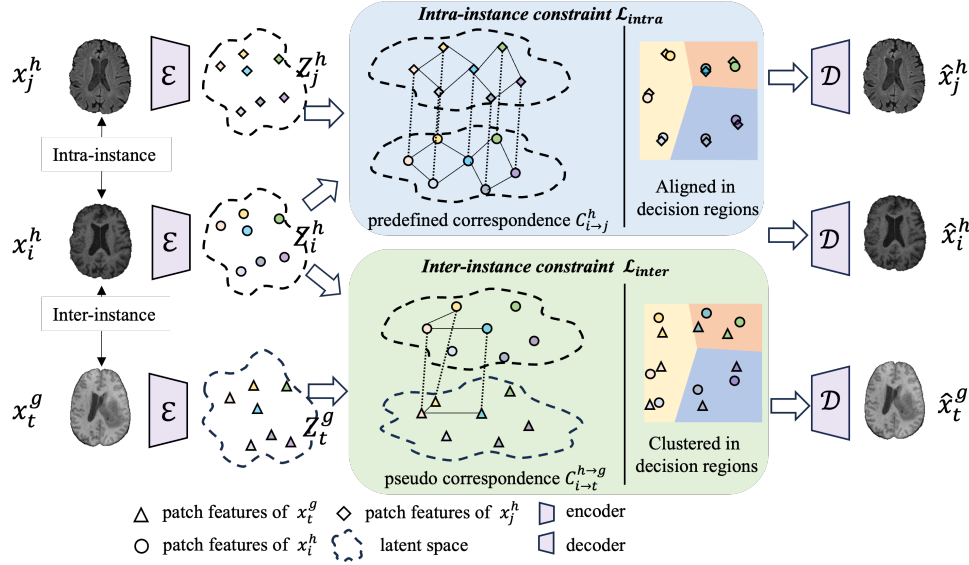


Figure 2. Pipeline of the proposed method. For each scan, a reconstruction loss learns unimodal representations. Across intra- and inter-individual pairs, following IM-NRC (Assumption 3.1), we align patch features of homologous anatomy, enforcing modality- and subject-invariant neighborhoods.

Z_j^h . To this end, we first obtain the pairwise distances in modality i on the feature level, which can be denoted as matrix D_i^h . Then, the neighborhood set $\mathcal{P}_{i,h}^\omega(a)$ of the a -th patch in Z_i^h can be defined as follows

$$\mathcal{P}_{i,h}^\omega(a) := \left\{ o \mid D_i^h[a, o] \in \text{Top}_\omega(D_i^h[a, :]), o \neq a, o \leq K \right\}, \quad (4)$$

where $\text{Top}_\omega(v)$ means the set of top- ω smallest values in vector v .

Given the correspondence $C_{i \rightarrow j}^h$, the set of positive features for $z_{j,a'}^h = C_{i \rightarrow j}^h(z_{i,a}^h)$ in modality j can be denoted as $\mathcal{P}'_{i \rightarrow j,h}(a) := \{o' \mid z_{j,o'}^h = C_{i \rightarrow j}^h(z_{i,o}^h), o \in \mathcal{P}_{i,h}^\omega(a)\}$, where the indices o are from the neighborhood set of the a -th patch in modality i .

Additionally, we define the negative set of a -th patch in Z_i^h as $\mathcal{N}_{i,h}(a) := \{o \mid o \leq K, o \notin \mathcal{P}_{i,h}^\omega(a), o \neq a\}$, which includes all patches except the ones in the neighborhood and the patch itself. In modality j , the corresponding negative set is $\mathcal{N}'_{i \rightarrow j,h}(a) := \{o' \mid o' = C_{i \rightarrow j}^h(o), o \in \mathcal{N}_{i,h}(a)\}$.

Hence, for a given p -th patch feature in Z_i^h , we can construct triplet sets in modality j for contrastive loss as follows

$$\text{Tri}_{i \rightarrow j, a, h}^{\text{intra}} := \left\{ (a', p', n') \mid a' = C_{i \rightarrow j}^h(a), (p', n') \in \mathcal{P}'_{i \rightarrow j, h}(a) \times \mathcal{N}'_{i \rightarrow j, h}(a) \right\}, \quad (5)$$

where $\mathcal{N}'_{i \rightarrow j, h}(a) \subset \mathcal{N}'_{i \rightarrow j, h}(a)$ is a randomly selected subset with ω samples (paired with top- ω positives). In the triplet set, (a', p') is the positive pair and (a', n') is the negative pair. Hence, to regularize the feature distribution in

Z_j^h with the relation in Z_i^h , we propose the following loss that aligns with Assumption 3.1,

$$\mathcal{L}_{\text{intra}}^{i \rightarrow j} = \frac{1}{|H| K \omega} \sum_{h \in H} \sum_{a \in Z_i^h} \sum_{\substack{(a', p', n') \\ \in \text{Tri}_{i \rightarrow j, a, h}^{\text{intra}}}} \left[d(z_{j,a'}^h, z_{j,p'}^h) - d(z_{j,a'}^h, z_{j,n'}^h) + \delta \right]_+, \quad (6)$$

where $[\cdot]_+ = \max(0, \cdot)$. Then, the final intra-instance loss is

$$\mathcal{L}_{\text{intra}} = \frac{1}{|M|(|M| - 1)} \sum_{i, j \in M} \mathcal{L}_{\text{intra}}^{i \rightarrow j}. \quad (7)$$

4.2. Inter-instance Constraint

The intra-instance loss described above is built on a prior one-to-one correspondence between tokens across modalities. However, in cross-subject settings, inter-individual anatomical and functional variability makes dense spatial alignment challenging. To overcome this limitation, we introduce a pseudo-correspondence for inter-instance constraints. This approach enforces local topological consistency only on partially matched token subsets, which are identified in an unsupervised manner as shown in Fig. 3.

Given two individual instances $h \in H$ and $g \in H$, let Z_i^h and Z_t^g denote the feature representations of $h \in H$ and $g \in H$ in modality i and t respectively. The pseudo correspondence between different instances is built based on mutual nearest neighborhood. Specifically, we define the correspondence $C_{i \rightarrow t}^{h \rightarrow g}$ under the distance metric $d(\cdot, \cdot)$ as

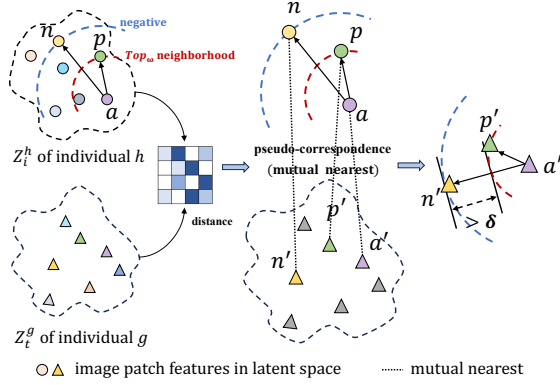


Figure 3. Inter-instance constraint. $\mathcal{L}_{inter}$ builds pseudo-correspondence by mutual-nearest-neighbor (MNN) and Top- n neighborhoods across patients, enforcing modality-invariant clustering within the same decision region. Intra-instance constraint can be a special case of Inter-instance constraint, where $h = g$.

follows

$$\begin{aligned} C_{i \rightarrow t}^{h \rightarrow g}(z_{i,a}^h) &= Z_{t,a'}^g \iff \\ a &= \arg \min_o d(z_{i,o}^h, Z_{t,a'}^g) \wedge a' = \arg \min_o d(z_{i,a}^h, Z_{t,o}^g). \end{aligned} \quad (8)$$

Then, we can build a set of solid indices according to the condition of the correspondence

$$\begin{aligned} V_{i,t}^{h,g} := \left\{ (a, a') \mid a = \arg \min_o d(z_{i,o}^h, Z_{t,a'}^g), \right. \\ \left. a' = \arg \min_o d(z_{i,a}^h, Z_{t,o}^g) \right\}. \end{aligned} \quad (9)$$

Given the a -th patch token feature $z_{i,a}^h \in Z_i^h$, similar to the Equation 5, we can define the triplet set for inter-instance loss

$$\begin{aligned} \text{Tri}_{i \rightarrow t, a, h \rightarrow g}^{\text{inter}} := \left\{ (a', p', n') \mid (a, a') \in V_{i,t}^{h,g}, \right. \\ \left. (p', n') \in \mathcal{P}_{i \rightarrow t, h \rightarrow g}'^\omega(a) \times \mathcal{N}_{i \rightarrow t, h \rightarrow g}'^\omega(a) \right\}. \end{aligned} \quad (10)$$

Thus, the inter-instance loss in Z_t^g with a -th patch in Z_i^h can be formulated as

$$\begin{aligned} \mathcal{L}_{inter}^{i \rightarrow t, h \rightarrow g}(a) &= \frac{1}{|\text{Tri}_{i \rightarrow t, a, h \rightarrow g}^{\text{inter}}|} \sum_{(a', p', n') \in \text{Tri}_{i \rightarrow t, a, h \rightarrow g}^{\text{inter}}} \\ &\quad \left[d(Z_{t,a'}^g, z_{t,p'}^g) - d(Z_{t,a'}^g, z_{t,n'}^g) + \delta \right]_+. \end{aligned} \quad (11)$$

Then, the loss for all patches is

$$\mathcal{L}_{inter}^{i \rightarrow t, h \rightarrow g} = \frac{1}{K} \sum_{a \in Z_i^h} \mathcal{L}_{inter}^{i \rightarrow t, h \rightarrow g}(a). \quad (12)$$

Then, the final inter-instance loss is

$$L_{inter} = \frac{1}{|M||H|(|M|-1)(|H|-1)} \sum_{i,t \in M} \sum_{h \in H} \mathcal{L}_{inter}^{i \rightarrow t, h \rightarrow g}. \quad (13)$$

Note that, if $V_{i,t}^{h,g} = \emptyset$, $L_{inter}^{i \rightarrow t, h \rightarrow g}$ will be set zero.

5. Experiments

Our experiments include evaluations on 4 segmentation tasks and 2 classification tasks. Beyond regular comparison with current SOTA medical SSL methods, we also compare missing modality segmentation ability with current SOTA missing-modality methods.

5.1. Experimental Setup

Pre-training Datasets. Following the multi-modal MRI pre-training protocol of BrainMVP (Rui et al., 2025), our model is pretrained on 16,022 3D scans from 3,755 patients. The corpus covers 8 modalities; for each patient, all modalities are registered before pre-training. As in BrainMVP, no segmentation annotations are used, and the pre-training data are disjoint from the downstream test sets. Details are shown in Tab 2, and additional details are provided in the Appendix.

Downstream datasets. Our downstream evaluation can be divided into three parts as shown in the Appendix Tab 2: 4 segmentation tasks (BraTS2023-PED (Kazerooni et al., 2024), BraTS2023-MET (Moawad et al., 2024), UPENN-GBM (Moawad et al., 2024)), and ISLES22 (Hernandez Petzsche et al., 2022), 2 classification tasks (ADNI (Jack Jr et al., 2008) and ADHD-200 (consortium, 2012)), and 1 missing modalities task (BraTS2018 segmentation (Menze et al., 2014)).

Implementation Details. Following prior medical SSL works (Wu et al., 2024; Wang et al., 2023c; Pan et al., 2025; Tan et al., 2025), we use Swin-B (Hatamizadeh et al., 2021) as the pre-training backbone. We reproduce all pre-training methods in Tab. 1 on the same pre-training dataset; the only exception is BrainMVP, for which we use the publicly released checkpoint pretrained on the same dataset. Pre-training runs for 10k iterations with batch size = 2, optimized by AdamW (Loshchilov & Hutter, 2017) with weight decay = 1e-5 and learning rate = 3e-4. The distance function $d(\cdot, \cdot)$ is the cosine distance. All pre-training and fine-tuning are conducted on a single NVIDIA A100. For downstream tasks, we select the checkpoint with the best validation performance and report results on the test set. Additional details are provided in the Appendix.

5.2. Evaluation on Downstream Tasks

5.2.1. COMPARISON WITH SOTA PRE-TRAINING METHODS

Since prior studies report that general-purpose SSL underperforms medical SSL on medical imaging, we restrict comparisons to medical pre-training baselines. For uni-modal methods, we include SwinMM (Wang et al., 2023c), Swin

Table 1. Results on four segmentation benchmarks. All methods are reproduced on the same pre-training dataset; for BrainMVP, we use its publicly released checkpoint trained on the same data. Metrics: TC (Tumor Core), ET (Enhancing Tumor), WT (Whole Tumor). **Best** and **second best** are highlighted. Our method achieves the best average performance.

Pre-training Method	Backbone	BraTS2023-PED				BraTS-MET				UPENN-GBM				ISLES22	All Dataset
		ET	TC	WT	avg.	ET	TC	WT	avg.	ET	TC	WT	avg.	IS	avg.
From Scratch															
-	UNET3D	46.12	84.92	86.63	72.55	53.47	58.24	63.97	58.56	80.36	87.14	85.04	84.18	77.81	73.28
-	UNETR	46.97	78.02	81.73	68.9	51.42	54.03	60.16	55.2	79.77	84.5	84.78	83.02	76.51	70.91
-	UniFormer	47.81	82.34	83.82	71.32	59.49	62.42	64.77	62.23	80.43	86.62	85.37	84.14	76.52	73.55
-	Swin UNETR	51.11	86.86	88.15	75.37	62.1	66.07	68.31	65.5	82.33	86.7	84.18	84.02	77.33	75.56
Medical SSL															
SwinMM (Wang et al., 2023c)	Swin UNETR	51.99	84.71	85.55	74.08	64.26	68.13	68.47	66.95	82.53	88.02	85.69	85.41	77.95	76.10
Swin UNETR (Tang et al., 2022)	Swin UNETR	52.55	84.75	86.28	74.53	63.32	67.5	69.42	66.74	81.84	87.36	85.91	85.04	78.71	76.26
VoCo (Wu et al., 2024)	Swin UNETR	50.52	84.22	85.91	73.55	60.51	64.13	66.01	63.55	81.92	87.7	86.02	85.21	79.65	75.49
M ³ AE (Liu et al., 2023)	UNET3D	47.57	84.62	85.78	72.66	57.73	62.97	67.31	62.67	81.98	88.11	85.68	85.26	79.13	74.93
BrainMVP (Rui et al., 2025)	UniFormer	49.89	85.25	87.52	74.22	62.42	67.63	70.45	66.83	81.81	87.65	84.67	84.71	79.69	76.36
S ² DC (Pan et al., 2025)	Swin UNETR	51.04	87.46	88.64	75.71	62.41	66.01	67.88	65.43	81.38	87.49	85.77	84.9	78.49	76.13
TACO	Swin UNETR	50.58	88.69	89.38	76.22	64.41	69.52	70.6	68.17	82.58	87.84	85.98	85.62	79.85	77.47

Table 2. The pre-training and downstream datasets in our work. Seg.: segmentation; cls: classification; ms: missing modality.

Dataset	Task Type	Modality type	Cases
Pre-training			
BraTS2021 (Baid et al., 2021)	-	T1,T1CE,T2,FLAIR	3755
BraTS2023-SSA (Adewole et al., 2023)	-	T1,T1CE,T2,FLAIR	75
BraTS2023-MEN (LaBella et al., 2023)	-	T1,T1CE,T2,FLAIR	1141
UCSF-PDGM (Calabrese et al., 2022)	-	T1,T1CE,T2,FLAIR,DWI,ADC	501
IXI (Chen et al., 2022)	-	T1,T2,MRA,PD	568
Downstream			
BraTS2023-PED (Kazerooni et al., 2024)	seg.	T1,T1CE,T2,FLAIR	99
BraTS2023-MET (Moawad et al., 2024)	seg.	T1,T1CE,T2,FLAIR	238
UPENN-GBM (Moawad et al., 2024)	seg.	T1,T1CE,FLAIR	127
ISLES22 (Hernandez Petzsche et al., 2022)	seg.	FLAIR,DWI,ADC	238
BraTS2018	seg. w/ ms.	T1,T1CE,T2,FLAIR	285
ADNI (Jack Jr et al., 2008)	cls.	T1	697
ADHD-200 (consortium, 2012)	cls.	T1	767

Table 3. Results on two classification benchmarks. All pre-training methods use the same pretrained model as in Tab. 1. **Best** and **second best** are highlighted. Our method achieves the best average performance in all metrics.

Pre-training Method	Backbone	ADNI			ADHD			avg.		
		ACC	AUC	F1	ACC	AUC	F1	ACC	AUC	F1
From scratch										
-	UNET3D	81.43	63.91	73.09	63.64	55.57	38.89	72.535	59.74	55.99
-	UniFormer	85.24	86.59	65.41	61.9	59.22	52.9	73.57	72.905	59.155
-	Swin-B	93.33	93.85	87.42	67.87	64.94	56.85	80.6	79.395	72.135
Medical SSL										
SwinMM	Swin-B	89.05	90.87	77.32	52.81	53.39	53.7	70.93	72.13	65.51
Swin UNETR	Swin-B	94.2	93.7	93.47	65.37	70.59	62.39	79.79	82.15	77.93
VoCo	Swin-B	85.51	90.3	82.23	53.68	52.43	48.06	69.60	71.37	65.15
M ³ AE	UNET3D	83.33	83.76	66.44	61.47	57.33	60.95	72.4	70.55	63.7
BrainMVP	UniFormer	93.33	95.05	87.1	69.26	68.72	58.41	81.3	81.89	72.76
S ² DC	Swin-B	88.57	95.2	76.67	64.94	70.34	63.06	76.76	82.77	69.87
TACO	Swin-B	96.19	97.09	93.57	67.53	71.14	63.82	81.86	84.12	78.7

UNETR (Tang et al., 2022), VoCo (Wu et al., 2024), and S²DC (Pan et al., 2025). For multi-modal pre-training, we compare against M³AE (Liu et al., 2023) and BrainMVP (Rui et al., 2025).

Results on segmentation tasks. Tab. 1 compares training from scratch with medical SSL pre-training across four benchmarks. The Dice similarity coefficient is adopted as the metric for evaluation. Training from scratch shows clear gaps to SSL in most tasks, e.g., the best scratch baseline (Swin UNETR) reaches a Dice coefficient 65.5% on BraTS-MET, 84.02% on UPENN-GBM, and 77.33% on ISLES22, below SSL counterparts. Our method, built on Swin UNETR backbone, delivers the best average on ev-

ery dataset and the best overall average across all datasets: 77.47% Dice, surpassing the strongest medical pre-training baseline (BrainMVP, 76.36%) by **1.11%** points. While some baselines win isolated sub-metrics (e.g., SwinMM on BraTS2023-PED ET, M³AE on UPENN-GBM TC), our approach yields consistent gains on the dataset-level averages, suggesting that the proposed pre-training better preserves transferable structure across cohorts and tasks.

Results on classification tasks. Tab. 3 reports accuracy (ACC), AUC, and F1 score on ADNI and ADHD datasets. Training from scratch is clearly inferior, particularly on the complex ADNI dataset. Incorporating medical SSL pre-training methods yields substantial improvements across various backbones, with BrainMVP and S²DC emerging as the strongest baselines, achieving best average scores of 81.3%/82.77%/69.87%, respectively. TACO establishes a new state of the art on both datasets, and all averaged metrics, reaching 81.86%/84.12%/78.7%, which yields gains of **+0.56%/+1.35%/+5.94%** over the best baseline. These consistent gains across datasets and metrics indicate that the proposed pre-training transfers better to classification than prior medical SSL methods, especially in terms of decision quality (F1) while maintaining strong discrimination (AUC).

5.2.2. MISSING MODALITY SEGMENTATION

Missing-modality setup. We follow the evaluation protocol of recent SOTA methods: PUIR (Tan et al., 2025), mmFormer (Zhang et al., 2022), SPA (Wang et al., 2023a), M³AE (Liu et al., 2023), and M2F (Shi et al., 2023). Following PUIR, we use the BraTS23 (Kazerooni et al., 2024) multi-modal brain tumor dataset, which provides four structural MRI modalities per subject (T1, T1ce, T2, FLAIR). We pretrain the models on BraTS23 and then fine-tune them on BraTS18 for segmentation, mirroring PUIR’s protocol; BraTS18 contains the same four modalities. The baseline results are from PUIR (Tan et al., 2025). More details can be found in the Appendix.

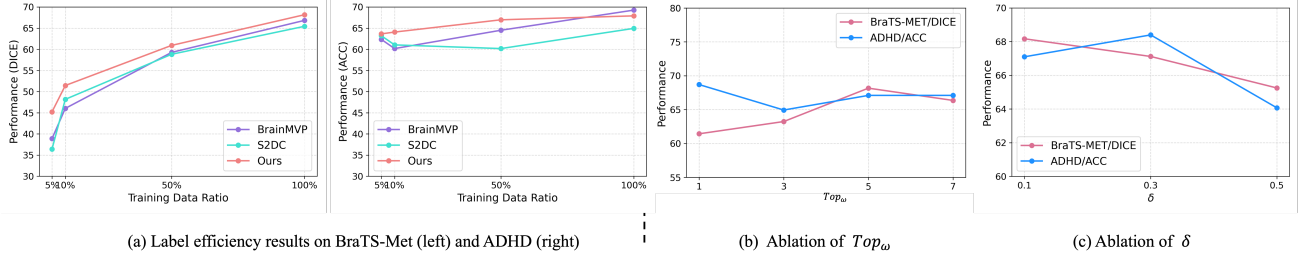


Figure 4. (a) Label efficiency results of the downstream segmentation task BraTS-MET and the classification task ADHD. Our method shows robust gains across data scales and tasks. (b) Ablation results of Top_{ω} . (c) Ablation results of δ .

Missing-modality robustness. Tab. 4 demonstrates the strong robustness of TACO under varying degrees of modality absence. Across the 15 evaluation settings covering TC, ET, and WT under FM and $MN \in 1, 2, 3$, TACO consistently ranks first or second, achieving the best performance in 10 settings. In terms of average performance over all modality configurations, TACO obtains the best Dice scores on ET (64.17%) and WT (89.55%), outperforming the strongest baseline PUIR by 0.68% and 1.92%, respectively, while remaining highly competitive on TC (79.68% vs. 79.78%). Notably, TACO achieves the best results on TC under FM, $MN=1$, and $MN=2$, and consistently delivers the highest WT performance across all missing-modality levels, including the most challenging $MN=3$ setting (86.50% vs. 85.17% for PUIR). For ET, TACO also achieves the best performance under $MN=2$ and remains the second-best method under FM, $MN=1$, and $MN=3$. These results indicate that TACO maintains consistently strong segmentation accuracy across different targets and missing-modality patterns, with particularly pronounced advantages for whole-tumor segmentation.

5.2.3. LABEL EFFICIENCY ANALYSIS

Fig. 4 shows performance as the labeled budget increases (we randomly sample 5%, 10%, 50%, and 100% of the training set). For the segmentation task, all methods improve with more data, but TACO consistently leads across budgets, with the largest margins in the low-label regime (5–10%), indicating stronger data efficiency. The gap narrows at 100%, yet our method remains best. For classification, TACO grows steadily from 5% to 100%, while S^2DC plateaus and BrainMVP even degrades at high budgets. Overall, our method provides the most robust gains across data scales and tasks.

5.3. Ablation Study

Ablation of losses. The results in the Table 5 demonstrate the effectiveness of incorporating both $\mathcal{L}_{inter}$ and $\mathcal{L}_{intra}$. Introducing either $\mathcal{L}_{inter}$ or $\mathcal{L}_{intra}$ leads to performance improvements. Overall, these results highlight that the proposed approach, by leveraging both intra-instance

and inter-instance topological consistency, enhances performance across both segmentation and classification tasks.

The impact of Top_{ω} . We investigate the impact of Top_{ω} in Equation 4 by pre-training models with $\omega \in [1, 3, 5, 7]$ and evaluating on BraTS-MET and ADHD. As shown in Fig. 4(b), performance on BraTS-MET improves steadily, peaking at Top_5 before slightly decreasing. Performance on ADHD peaks at $\omega = 3$ and declines thereafter. Based on these results, we select $\omega = 5$ for subsequent experiments to balance both tasks.

Table 4. Missing-modality segmentation results on BraTS18. MN denotes the number of missing modalities, and FM denotes full modalities. All settings reports statistics over all modality settings, including FM. Mean/Std. denote the mean and sample standard deviation over the corresponding modality combinations. Detailed results are provided in the Appendix.

Method	All settings		FM –	MN=1		MN=2		MN=3	
	Mean	Std.		Mean	Std.	Mean	Std.	Mean	Std.
Tumor Core									
RFNET	76.08	6.99	83.40	80.63	4.53	76.57	5.75	68.95	6.23
mmFormer	76.43	5.83	82.22	79.78	4.33	76.55	5.35	71.45	5.75
SPA	74.80	6.95	82.23	78.99	5.38	75.01	6.16	68.44	6.15
M ³ AE	72.67	7.43	80.29	77.61	4.59	73.37	5.68	64.79	6.87
M2F	73.69	6.83	80.34	77.48	5.19	74.17	6.07	67.51	6.63
BrainMVP	74.32	6.91	83.83	78.97	5.95	73.33	5.24	68.79	6.33
PUIR	79.78	4.55	86.72	83.64	2.29	79.56	2.83	74.51	2.22
TACO	79.68	5.98	87.45	84.32	3.58	80.10	3.89	72.48	3.26
Enhancing Tumor									
RFNET	59.31	15.10	73.65	66.91	12.90	59.17	14.98	48.35	14.74
mmFormer	62.14	18.60	79.91	71.54	15.94	61.77	18.48	48.86	18.23
SPA	58.92	17.68	73.40	68.44	17.11	58.05	17.43	47.10	16.77
M ³ AE	55.98	17.45	73.79	65.09	14.54	55.53	16.57	43.09	17.72
M2F	58.84	16.58	75.26	66.67	13.83	58.99	16.33	46.70	16.71
BrainMVP	58.48	13.24	73.64	62.25	10.70	58.23	17.06	51.29	7.32
PUIR	63.49	7.58	70.64	64.44	6.62	63.87	6.66	60.19	10.85
TACO	64.17	12.04	75.31	69.44	11.20	64.05	11.97	56.26	12.28
Whole Tumor									
RFNET	83.92	6.14	89.27	87.25	2.94	84.95	5.24	77.70	6.60
mmFormer	84.84	5.35	88.26	87.59	2.40	85.36	5.08	80.45	6.67
SPA	84.52	5.48	89.03	87.81	1.25	85.26	4.16	78.98	6.81
M ³ AE	81.52	6.71	86.82	85.64	1.36	82.43	5.30	74.74	8.16
M2F	83.88	5.79	88.72	87.30	1.99	84.62	4.52	78.13	7.05
BrainMVP	82.85	8.48	89.83	88.78	0.83	82.43	7.57	75.82	10.50
PUIR	87.63	3.25	91.19	89.49	1.82	87.45	3.01	85.17	3.74
TACO	89.55	3.25	92.56	91.51	1.12	89.78	2.68	86.50	3.92

The impact of δ . Fig. 4 (c) shows the impact of the margin δ in $\mathcal{L}_{intra}$ and $\mathcal{L}_{inter}$. For BraTS-MET, performance declines with increasing δ , while ADHD performs best at $\delta = 0.3$. To balance both tasks, we select $\delta = 0.3$.

Table 5. Ablation results of L_{intra} and L_{inter} .

Baseline(L_{uni})	L_{intra}	L_{inter}	BraTS-MET				ADHD		
			ET	TC	WT	avg.	ACC	AUC	F1
•			58.24	61.54	65.10	61.63	64.94	65.17	63.06
•	•		64.63	68.50	69.74	67.62	64.94	67.91	57.61
•		•	63.94	68.20	69.05	67.07	65.8	70.60	61.44
•	•	•	64.41	69.52	70.6	68.17	67.53	71.14	63.82

5.4. Analysis of IM-NRC

Inter-patient embedding structure (Fig. 5). We visualize patch embeddings from four patients as shown in Fig. 5. Features from a randomly initialized model collapse into a single amorphous cloud, indicating no meaningful structure. BrainMVP forms a few large clusters dominated by a single color, i.e., features are separated mainly by patient identity, suggesting patient-specific memorization that can hurt cross-patient transfer. S^2DC produces many fragmented clusters with mixed colors, while some local separability exists, but the manifold is noisy and lacks global organization. TACO yields a compact distribution, with repeatedly occurring clusters where colors are well interleaved. This pattern indicates modality/instance-invariant alignment and neighborhood consistency across instances, supporting the observed gains in both segmentation and classification.

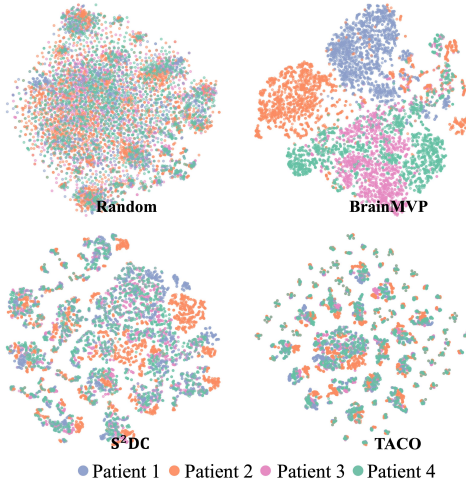


Figure 5. The t-SNE visualization of token-level representations across patients. TACO achieves clear and well-organized clusters across patients, demonstrating superior cross-instance learning.

Intra-patient multi-modal embedding structure (Fig. 6).

We color points by modality as shown in Fig. 6. For patches from two modalities in the same patient, grounded patch correspondences are found, with aligned pairs highlighted by grey lines. Random embeddings collapse into a noisy cloud with no clear structure. BrainMVP shows a clear modality split (orange vs. green) with many long inter-cluster links, indicating modality-specific embeddings and hindering cross-modal transfer. S^2DC forms modality-dominated islands, with frequent edge crossings that reveal unstable neighborhoods. In contrast, TACO produces compact, well-

separated micro-clusters with interleaved colors and mostly intra-cluster edges, demonstrating modality-invariant alignment and stable neighborhood consistency, which supports its robustness in downstream tasks and missing-modality settings.

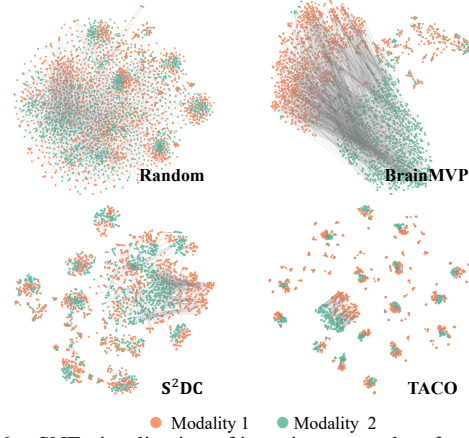


Figure 6. t-SNE visualization of intra-instance token feature alignment across 2 modalities. “Random” refers to a model with random initialization. TACO shows superior alignment, with more coherent clusters.

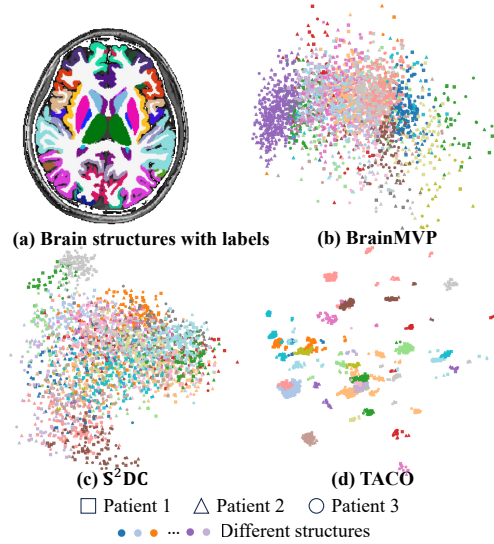


Figure 7. Visualization of brain structure representations across methods. (a) Ground-truth labels of brain structures, with 10 structures selected for the visualizations in (b)(c)(d) methods. TACO shows superior performance in region clustering.

Clustering on Brain Tissues (Fig. 7). We generate brain structure labels for 3 MRI T1 volumes from 3 individuals in the Queensland Twin Adolescent Brain (QTAB) dataset (Strike et al., 2023) using the tool (Billot et al., 2023). Using pixel-level structure labels, Fig. 7 presents token-level visualizations of brain structures across different methods. In Fig. 7(a), the ground-truth labels show the expected anatomical segmentation. The representations from BrainMVP and S^2DC exhibit some clustering

of patient-specific features, but a less distinct separation between anatomical regions is observed. TACO provides the most coherent representation, with well-defined clusters corresponding to both anatomical regions and patient identities. Our method achieves the best cross-patient and cross-structure alignment, demonstrating superior generalization across individuals and structures.

6. Conclusion

This work introduces a novel approach to learning topological consistency in brain structures across instances, extending beyond traditional instance-level self-supervision through the proposed IM-NRC assumption. Extensive evaluation on seven benchmark datasets, along with visualizations of patch features, further highlights the efficacy of our approach. **Limitation.** Future work can expand TACO to include more modalities and anatomical structures, such as whole-body PET-CT imaging, to increase its applicability.

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Impact Statement

This paper presents a method to enhance generalization ability across instances/people and modalities in medical self-supervised learning, which is crucial for foundation-modeling in medical imaging. This could potentially reduce reliance on expensive expert annotations, improve data efficiency, and enable more robust downstream performance in clinically relevant tasks such as segmentation, detection, and disease classification, particularly in settings where labeled data are scarce or heterogeneous across institutions.

References

- Adewole, M., Rudie, J. D., Gbdamosi, A., Toyobo, O., Raymond, C., Zhang, D., Omidiji, O., Akinola, R., Suwaid, M. A., Emegoakor, A., et al. The brain tumor segmentation (brats) challenge 2023: Glioma segmentation in sub-saharan africa patient population (brats-africa). *ArXiv*, pp. arXiv–2305, 2023. 6
- Baid, U., Ghodasara, S., Mohan, S., Bilello, M., Calabrese, E., Colak, E., Farahani, K., Kalpathy-Cramer, J., Kitamura, F. C., Pati, S., et al. The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv preprint arXiv:2107.02314*, 2021. 6
- Billot, B., Greve, D. N., Puonti, O., Thielscher, A., Van Leemput, K., Fischl, B., Dalca, A. V., Iglesias, J. E., et al. Synthseg: Segmentation of brain mri scans of any contrast and resolution without retraining. *Medical image analysis*, 86:102789, 2023. 8
- Calabrese, E., Villanueva-Meyer, J. E., Rudie, J. D., Rauschecker, A. M., Baid, U., Bakas, S., Cha, S., Mongan, J. T., and Hess, C. P. The university of california san francisco preoperative diffuse glioma mri dataset. *Radiology: Artificial Intelligence*, 4(6):e220058, 2022. 6
- Chen, C., Dou, Q., Jin, Y., Liu, Q., and Heng, P. A. Learning with privileged multimodal knowledge for unimodal segmentation. *IEEE transactions on medical imaging*, 41(3):621–632, 2021. 2
- Chen, J., Frey, E. C., He, Y., Segars, W. P., Li, Y., and Du, Y. Transmorph: Transformer for unsupervised medical image registration. *Medical Image Analysis*, 2022. 6
- Chen, X., Fan, H., Girshick, R., and He, K. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020. 2
- consortium, A.-. The adhd-200 consortium: a model to advance the translational potential of neuroimaging in clinical neuroscience. *Frontiers in systems neuroscience*, 6:62, 2012. 5, 6
- Ding, Y., Yu, X., and Yang, Y. Rfnet: Region-aware fusion network for incomplete multi-modal brain tumor segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 3975–3984, 2021. 2
- Elsayed, N., Osman, Y. B. M., Li, C., Liu, J., Si, W., Zhang, J., and Wang, S. Automating vessel segmentation in the heart and brain: A trend to develop multi-modality and label-efficient deep learning techniques. *Biomedical Signal Processing and Control*, 109:108028, 2025. 3
- Fischl, B. Freesurfer. *Neuroimage*, 62(2):774–781, 2012. 3
- Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P., Buchatskaya, E., Doersch, C., Avila Pires, B., Guo, Z., Gheshlaghi Azar, M., et al. Bootstrap your own latent: a new approach to self-supervised learning. *Advances in neural information processing systems*, 33:21271–21284, 2020. 1
- Hatamizadeh, A., Nath, V., Tang, Y., Yang, D., Roth, H. R., and Xu, D. Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images. In *International MICCAI brainlesion workshop*, pp. 272–284. Springer, 2021. 2, 5

- He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9729–9738, 2020. 1
- He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 16000–16009, 2022. 1
- He, Y., Yang, G., Ge, R., Chen, Y., Coatrieux, J.-L., Wang, B., and Li, S. Geometric visual similarity learning in 3d medical image self-supervised pre-training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9538–9547, 2023. 2, 1
- Hernandez Petzsche, M. R., De La Rosa, E., Hanning, U., Wiest, R., Valenzuela, W., Reyes, M., Meyer, M., Liew, S.-L., Kofler, F., Ezhov, I., et al. Isles 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset. *Scientific data*, 9(1):762, 2022. 5, 6
- Hu, X., Li, F., Samaras, D., and Chen, C. Topology-preserving deep image segmentation. *Advances in neural information processing systems*, 32, 2019. 3
- Iglesias, J. E., Billot, B., Balbastre, Y., Magdamo, C., Arnold, S. E., Das, S., Edlow, B. L., Alexander, D. C., Golland, P., and Fischl, B. Synths: A public ai tool to turn heterogeneous clinical brain scans into high-resolution t1-weighted images for 3d morphometry. *Science advances*, 9(5):eadd3607, 2023. 2
- Jack Jr, C. R., Bernstein, M. A., Fox, N. C., Thompson, P., Alexander, G., Harvey, D., Borowski, B., Britson, P. J., L. Whitwell, J., Ward, C., et al. The alzheimer’s disease neuroimaging initiative (adni): Mri methods. *Journal of Magnetic Resonance Imaging: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 27(4):685–691, 2008. 5, 6
- Jiang, Y., Sun, M., Guo, H., Bai, X., Yan, K., Lu, L., and Xu, M. Anatomical invariance modeling and semantic alignment for self-supervised learning in 3d medical image analysis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15859–15869, 2023. 1, 2
- Kazerooni, A. F., Khalili, N., Liu, X., Haldar, D., Jiang, Z., Anwar, S. M., Albrecht, J., Adewole, M., Anazodo, U., Anderson, H., et al. The brain tumor segmentation (brats) challenge 2023: Focus on pediatrics (cbtnc-connect-dipgr-asnr-miccai brats-peds). *ArXiv*, pp. arXiv–2305, 2024. 5, 6
- LaBella, D., Adewole, M., Alonso-Basanta, M., Altes, T., Anwar, S. M., Baid, U., Bergquist, T., Bhalerao, R., Chen, S., Chung, V., et al. The asnr-miccai brain tumor segmentation (brats) challenge 2023: Intracranial meningioma. *arXiv preprint arXiv:2305.07642*, 2023. 6
- Liu, H., Wei, D., Lu, D., Sun, J., Wang, L., and Zheng, Y. M3ae: multimodal representation learning for brain tumor segmentation with missing modalities. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 1657–1665, 2023. 6
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 5
- Mao, C., Zhang, L., Joshi, A., Yang, J., Wang, H., and Vondrick, C. Robust perception through equivariance. *arXiv preprint arXiv:2212.06079*, 2022. 3
- Menze, B. H., Jakab, A., Bauer, S., Kalpathy-Cramer, J., Farahani, K., Kirby, J., Burren, Y., Porz, N., Slotboom, J., Wiest, R., et al. The multimodal brain tumor image segmentation benchmark (brats). *IEEE transactions on medical imaging*, 34(10):1993–2024, 2014. 5
- Moawad, A. W., Janas, A., Baid, U., Ramakrishnan, D., Saluja, R., Ashraf, N., Maleki, N., Jekel, L., Yordanov, N., Fehringer, P., et al. The brain tumor segmentation-metastases (brats-mets) challenge 2023: Brain metastasis segmentation on pre-treatment mri. *ArXiv*, pp. arXiv–2306, 2024. 5, 6
- Pan, T., Tan, Z., Guo, K., Xu, D., Xu, W., Jiang, C., Guo, X., Qi, Y., and Cheng, Y. Structure-aware semantic discrepancy and consistency for 3d medical image self-supervised learning. *arXiv preprint arXiv:2507.02581*, 2025. 1, 2, 5, 6
- Rui, S., Chen, L., Tang, Z., Wang, L., Liu, M., Zhang, S., and Wang, X. Multi-modal vision pre-training for medical image analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 5164–5174, 2025. 1, 2, 5, 6
- Segobin, S., Haast, R. A., Kumar, V. J., Lella, A., Alkemade, A., Bach Cuadra, M., Barbeau, E. J., Felician, O., Pergola, G., Pitel, A.-L., et al. A roadmap towards standardized neuroimaging approaches for human thalamic nuclei. *Nature Reviews Neuroscience*, 25(12):792–808, 2024. 2
- Shi, J., Yu, L., Cheng, Q., Yang, X., Cheng, K.-T., and Yan, Z. Mftrans: Modality-masked fusion transformer for incomplete multi-modality brain tumor segmentation. *IEEE journal of biomedical and health informatics*, 28(1):379–390, 2023. 6

- Strike, L. T., Hansell, N. K., Chuang, K.-H., Miller, J. L., de Zubicaray, G. I., Thompson, P. M., McMahon, K. L., and Wright, M. J. The queensland twin adolescent brain project, a longitudinal study of adolescent brain development. *Scientific Data*, 10(1):195, 2023. [8](#)
- Tan, Z., Yang, X., Pan, T., Liu, T., Jiang, C., Guo, X., Wang, Q., Nguyen, A., Qi, Y., Huang, K., et al. Towards a universal 3d medical multi-modality generalization via learning personalized invariant representation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 21895–21905, 2025. [1](#), [2](#), [5](#), [6](#)
- Tang, Y., Yang, D., Li, W., Roth, H. R., Landman, B., Xu, D., Nath, V., and Hatamizadeh, A. Self-supervised pre-training of swin transformers for 3d medical image analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 20730–20740, 2022. [2](#), [6](#)
- Wald, T., Ulrich, C., Suprijadi, J., Ziegler, S., Nohel, M., Peretzke, R., Kohler, G., and Maier-Hein, K. An open-mind for 3d medical vision self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 23839–23879, 2025. [2](#)
- Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L., and Carneiro, G. Multi-modal learning with missing modality via shared-specific feature modelling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15878–15887, 2023a. [6](#)
- Wang, S., Yan, Z., Zhang, D., Wei, H., Li, Z., and Li, R. Prototype knowledge distillation for medical segmentation with missing modality. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5. IEEE, 2023b. [2](#)
- Wang, Y., Zhang, Y., Liu, Y., Lin, Z., Tian, J., Zhong, C., Shi, Z., Fan, J., and He, Z. Acn: adversarial co-training network for brain tumor segmentation with missing modalities. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VII 24*, pp. 410–420. Springer, 2021. [2](#)
- Wang, Y., Li, Z., Mei, J., Wei, Z., Liu, L., Wang, C., Sang, S., Yuille, A. L., Xie, C., and Zhou, Y. Swinmm: masked multi-view with swin transformers for 3d medical image segmentation. In *International conference on medical image computing and computer-assisted intervention*, pp. 486–496. Springer, 2023c. [2](#), [5](#), [6](#)
- Wen, B., Zhang, H., Bartsch, D.-U. G., Freeman, W., Nguyen, T., and An, C. Topology-preserving image segmentation with spatial-aware persistent feature matching. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 5762–5771, 2025. [3](#)
- Wu, L., Zhuang, J., and Chen, H. Voco: A simple-yet-effective volume contrastive learning framework for 3d medical image analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22873–22882, 2024. [2](#), [5](#), [6](#)
- Zhang, Y., He, N., Yang, J., Li, Y., Wei, D., Huang, Y., Zhang, Y., He, Z., and Zheng, Y. mmformer: Multimodal medical transformer for incomplete multimodal learning of brain tumor segmentation. In *International conference on medical image computing and computer-assisted intervention*, pp. 107–117. Springer, 2022. [2](#), [6](#)
- Zhao, Z., Yang, H., and Sun, J. Modality-adaptive feature interaction for brain tumor segmentation with missing modalities. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 183–192. Springer, 2022. [2](#)

A. Explanation of Concepts

A.1. Symbols and Denotations

Tab. 10 shows the symbols and the corresponding denotations.

A.2. IM-NRC

The Instance-agnostic Cross-Modal Neighborhood Ranking Consistency (IM-NRC) assumption proposes a phenomenon that consistently occurs in physiology. It is built upon the concept of topological consistency, which refers to the inherent connectivity and structure of an object. In the human body, especially the brain, physiological structures are naturally interconnected in ways that cannot be altered without fundamentally changing their organization. Moreover, this connectivity is preserved not only during rigid transformations but also under non-rigid deformations. Thus, we adopt this relational position consistency rather than relying on strictly constrained patch alignment. Fig. 8a shows this motivation.

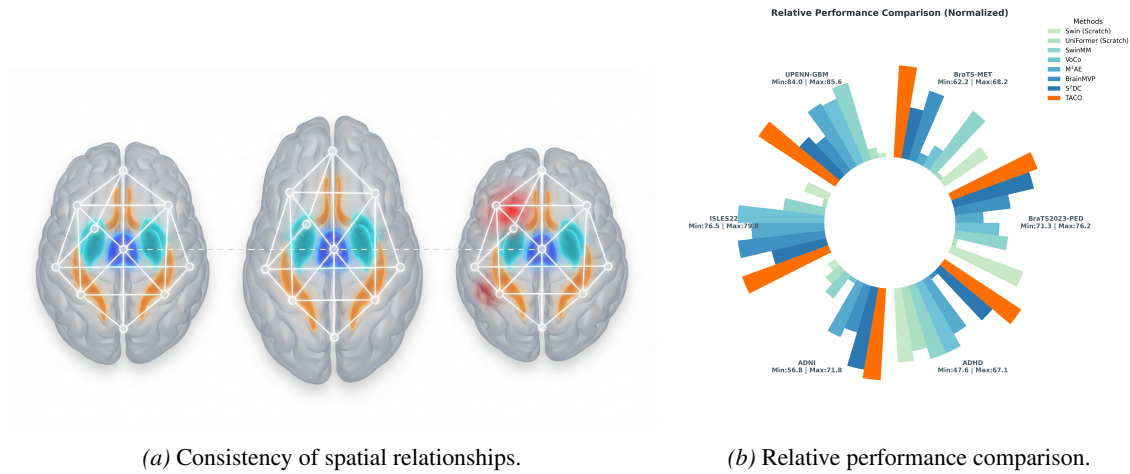


Figure 8. Visualization of (a) cross-subject spatial consistency and (b) normalized relative performance metrics.

B. Comparison with SOTA

Fig. 8b presents a relative performance comparison with min-max scaling normalization. From the figure, it is evident that TACO exhibits standout performance across all datasets.

Compared with GVSL (He et al., 2023). GVSL maintains the geometric consistency with predictable explicit transformation. Specifically, the model predicts the affine matrix and deformable mappings between two images. As illustrated in GVSL, the motivation and techniques aim to cluster features with the same semantics across different images, which is fundamentally different from ours. First, as indicated by the title and problem formulation, our method considers the multi-modal data rather than two augmented views derived from a single modality. The transformation between different modalities can not be explicitly presented. Second, we do not directly optimize triplet distances across two views/modalities. Instead, we define triplets in one modality and apply the corresponding ranking/metric loss in the other modality, thereby enforcing cross-modal geometric consistency of the representation space. Direct clustering of features across images/views can preserve only global organ-structure consistency within one modality or instance, while overlooking the conformal correspondences required for robust cross-modality or cross-instance alignment. These objectives and mechanisms are therefore fundamentally different from GVSL. **GVSL results are not included in our tables for the following reasons.** According to the official GVSL repository, the method is not intended to be applied to MRI data. In our own attempts to reproduce GVSL under the MRI setting, we also encountered consistent convergence issues, which prevented us from obtaining stable and reliable results for a fair comparison.

C. Robustness to registration error

We simulate rigid registration residuals by applying random translations/rotations to one modality only, while keeping the other modality unchanged, and then re-evaluate the latent space. The performance is in the following table:

Table 6. Robustness under different levels of rigid errors.

Setting	Pos. cos dist	Top-1 retrieval	MNN selected ratio
Clean	0.0175	93.31%	89.75%
Mild rigid error (max 2 vox / 2°)	0.0184	92.98%	88.96%
Moderate error (max 4 vox / 5°)	0.0209	91.55%	86.61%
Strong error (max 8 vox / 10°)	0.0277	86.59%	79.06%

D. Distance between positive and negative tokens

We provide quantitative latent-space disentanglement results of our method across 3,396 pretraining subjects. The hard negative point means the nearest negative point to the anchor point. The performance illustrates the strong separation between positive and negative samples, as shown in the following table:

Table 7. Distance between positive and negative token.

Metric	Value
Pos cos dist	0.0175 ± 0.0004
Neg cos dist	0.5502 ± 0.0003
Hard-neg cos dist	0.0616 ± 0.0002
Neg-Pos gap	0.5327
Hard-neg-Pos gap	0.0441
Top-1 retr. acc.	$93.31\% \pm 0.26\%$
Top-5 retr. acc.	$99.27\% \pm 0.06\%$
Pairwise rank acc.	$99.45\% \pm 0.03\%$

E. Computational Complexity

We compare the computational complexity with Swin-B-based pretrained models from the number of parameters and TFLOPS, as Tab. 8 shows. All inputs are set as $96 \times 96 \times 96$.

Table 8. A comparison of computational complexity

Methods	Parameters	TFLOPS
VoCo	127.44M	15.48
S ² DC	63.93M	5.09
TACO	39.96M	4.40

From the open-source code, VoCo, which uses both a teacher and a student model, includes 5 extension heads for different outputs from the Swin-B encoder. This setup introduces additional parameters and increases the TFLOPS during pre-training. Similarly, S²DC also employs a teacher-student model structure, adding extra parameters. In contrast, TACO, which only has a small decoder head and does not include teacher-student structures, demonstrates the best performance in terms of parameters and TFLOPS.

F. Experiments on OpenMind

To ensure a rigorous evaluation, we integrated our framework into the SSL3D OpenMind ecosystem. We evaluated TACO against these SOTA baselines across two segmentation datasets, ISLES and MSLS2017. The reported evaluation metrics are DSCO(Dice Similarity Coefficient) and NSD(Normalized Surface Distance). We provide both the average and the standard deviation value. The results are provided at Table 9.

Table 9. Comparison of pre-training methods on ISLES and MSLS2017.

Pre-training method	ISLES				MSLS2017			
	DSC Avg.	DSC Std.	NSD Avg.	NSD Std.	DSC Avg.	DSC Std.	NSD Avg.	NSD Std.
MAE (best in OpenMind)	74.02	2.38	71.82	0.85	83.02	4.46	92.66	4.82
TACO (Ours)	75.34	2.44	72.12	1.28	83.96	4.50	93.50	4.81

G. Implementation Details

All code, including pre-training, fine-tuning, and data splits, will be made publicly available upon publication.

Table 10. Summary of notations used throughout the paper.

Symbol	Description
i, j, t	Modality categories
a, p, n, k	Patch indices
h, g	Individual IDs
x_i^h	Image data from individual h in modality i
$\mathcal{E}$	Transformer encoder
$\mathcal{D}$	Decoder
Z_i^h	Output tokens from x_i^h through $\mathcal{E}$
K	Output token number from x_i^h through $\mathcal{E}$
$z_{i,k}^h$	The k -th token in Z_i^h
$C_{i \rightarrow j}^{h \rightarrow g}$	Correspondence from modality i to j and from individual h to g
$C_{i \rightarrow j}^h$	Correspondence from modality i to j within individual h
$\mathcal{P}_{i,h}^\omega(a)$	The neighborhood set of ω nearest patches' indices in Z_i^h at a -th patch.
$\mathcal{N}_{i,h}(a)$	The non-neighborhood set of with respect to $\mathcal{P}_{i,h}^\omega(a)$.
$\mathcal{P}_{i,h}^{\prime\omega}(a), \mathcal{P}_{i \rightarrow t, h \rightarrow g}^{\prime\omega}(a)$	The set of corresponding patch indices with respect to the neighborhood set.
$\mathcal{N}_{i,h}^{\prime\omega}(a), \mathcal{N}_{i \rightarrow t, h \rightarrow g}^{\prime\omega}(a)$	The set of corresponding patch indices with respect to the non-neighborhood set.
$\text{Tri}_{i \rightarrow j, a, h}^{\text{intra}}, \text{Tri}_{i \rightarrow t, a, h \rightarrow g}^{\text{inter}}$	The set of triplet patches' indices at a -th patch.
$V_{i,t}^{h,g}$	The set of solid indices pairs for anchor points selection.
$\mathcal{L}_{\text{uni}}, \mathcal{L}_{\text{intra}}, \mathcal{L}_{\text{inter}}$	Loss functions used in our method.

G.1. Pre-training

For all reproduced open-source methods, we use their official implementations. For TACO, we utilize the Swin-B model for pretraining, paired with a 6-layer CNN-UPSAMPLE-InstanceNorm3d decoder. The hyperparameters are set as follows: learning rate= $3e-4$, weight decay= $1e-5$, iteration=10k, optimizer=AdamW, and input volume size= $96 \times 96 \times 96$. Additionally, we apply a cosine annealing scheduler with warmup.

G.2. Fine-tuning

All methods, including those trained from scratch, are evaluated under consistent fine-tuning settings. For multi-modal tasks, we follow the same strategy as BrainMVP, initializing the encoder with pre-trained weights, except for the patch embedding.

Segmentation tasks. Swin-B-based pre-trained models follow the Swin UNETR structure, ViT-B-based pre-trained models follow the UNETR structure, and UniFormer-based pre-trained models follow the UniUNET structure (as in BrainMVP). For M³AE, the UNET3D structure is used. The hyperparameters are set as follows: learning rate= $1e-4$, weight decay= $1e-5$, epochs=300, optimizer=AdamW, and input volume size= $96 \times 96 \times 96$. Except for ISLES, we adopt input volume size= $128 \times 128 \times 128$. During inference, a sliding window size of 0.75 and the same input volume size are used. A cosine annealing scheduler with warmup is also applied. The data is split as 6:1:3 for training, validation, and testing, respectively, with the model yielding the best validation result being selected for evaluation on the test dataset.

Classification tasks. For those tasks, all pre-trained models adopt their encoder (i.e., Swin-B, ViT-B, UniFormer, and 3D CNN, respectively) for classification. The hyperparameters are set as follows: learning rate= $1e-3$, weight decay= $1e-5$,

Table 11. The results of missing modalities. • represents the missed modality.

Method	MN=3				MN=2						MN=1				FM
FLAIR	•	•	•	•			•	•		•			•		
T1	•			•	•				•			•			
T1ce	•	•		•	•		•		•						
T2	•	•	•		•	•		•			•				
Tumor Core															
RFNET	65.86	65.27	78.26	66.40	72.99	83.23	70.66	81.25	70.66	80.63	83.22	73.89	83.36	82.05	83.40
mmFormer	69.40	65.45	79.12	71.84	79.90	70.45	82.79	81.17	71.62	73.35	81.78	82.42	73.31	81.61	82.22
SPA	67.80	77.32	64.56	64.08	81.51	79.43	69.14	70.63	68.60	80.75	81.75	70.92	81.74	81.55	82.23
M ³ AE	64.03	74.53	58.63	61.95	79.20	77.45	69.25	67.48	67.98	78.85	80.15	70.75	79.40	80.15	80.29
M2F	65.79	63.29	77.31	63.64	70.38	79.93	68.01	79.62	67.68	79.37	80.65	69.73	80.01	79.53	80.34
PUIR	75.83	71.20	75.29	75.71	80.66	83.60	79.23	74.83	79.51	79.52	83.92	82.78	86.65	81.22	86.72
TACO	71.65	75.07	68.22	74.99	86.09	78.14	83.64	77.12	79.19	76.40	86.62	87.36	79.45	83.88	87.45
Enhancing tumor															
RFNET	39.85	41.39	70.43	41.72	45.99	73.07	45.25	72.87	45.25	72.59	73.52	47.56	73.01	73.55	73.65
mmFormer	37.00	38.41	75.80	44.22	78.09	45.20	79.36	78.16	41.71	48.12	79.14	80.06	47.63	79.31	79.91
SPA	40.08	72.19	38.89	37.23	73.11	73.06	40.64	42.27	43.65	75.56	43.34	81.74	73.36	75.31	73.40
M ³ AE	38.69	69.22	30.89	33.56	71.40	70.90	38.53	41.91	40.90	69.51	71.61	43.37	71.17	74.20	73.79
M2F	37.99	37.79	71.74	39.28	43.37	74.66	45.42	73.48	43.50	73.48	73.56	45.93	73.15	74.03	75.26
PUIR	67.45	54.83	70.86	47.63	69.38	52.91	70.10	59.45	67.44	63.91	70.79	57.78	69.42	59.76	70.64
TACO	47.16	74.34	50.61	52.96	74.74	51.76	74.44	75.66	53.35	54.39	75.34	74.75	52.65	75.02	75.31
Whole tumor															
RFNET	85.77	72.69	71.95	80.40	87.82	87.97	88.27	75.57	88.27	81.80	88.30	88.78	89.06	82.87	89.27
mmFormer	87.78	74.69	74.91	84.43	76.09	84.48	89.63	84.40	88.64	88.91	84.04	89.29	88.58	88.45	88.26
SPA	84.09	72.85	73.37	85.60	85.97	76.93	87.09	86.09	87.55	87.94	88.36	88.16	88.74	85.96	89.03
M ³ AE	80.52	67.06	68.42	82.96	82.57	71.97	85.82	83.25	86.00	84.94	86.06	86.53	86.34	83.61	86.82
M2F	85.72	72.48	71.78	82.53	87.73	87.66	84.35	76.03	87.69	84.27	88.17	88.22	88.47	84.32	88.72
PUIR	89.23	81.73	82.26	87.45	89.74	89.03	88.00	81.92	89.72	86.27	89.12	90.50	91.25	87.10	91.19
TACO	91.23	83.40	83.15	88.22	91.51	91.69	89.78	84.86	91.87	88.99	91.79	92.54	91.79	89.92	92.56

epochs=50, optimizer=Adam, and input volume size= $128 \times 128 \times 128$. A cosine annealing scheduler is applied. Same as segmentation, the datasets are split as 6:1:3 for training, validation, and testing, respectively, with the model yielding the best validation result being selected for evaluation on the test dataset.

G.3. Pre-training Datasets

BraTS2021 Multimodal Brain Tumor Segmentation Challenge focuses on glioma segmentation. BraTS2021 includes T1, T1CE, T2, and FLAIR modalities with 1470 patients.

BraTS2023-SSA is glioma segmentation in sub-saharan Africa patient population (brats-africa), including T1, T1CE, T2, and FLAIR modalities with 75 patients.

BraTS2023-MEN is a meningioma segmentation challenge with 1141 patients.

UCSF-PDGM UCSF-PDGM is a publicly released preoperative MRI dataset of diffuse glioma, jointly developed by the University of California San Francisco (UCSF) and the University of Pennsylvania (UPenn). It includes 501 patients with T1, T1CE, T2, FLAIR, DWI, and ADC modalities.

IXI Our MRI dataset of IXI is extracted from Information eXtraction from Images. To keep the same as BrainMVP, we also select 568 patients. It has T1, T2, MRA, and PD modalities.

G.4. Downstream Datasets

BraTS2023-PED. The BraTS2023-PEDs dataset is a multi-institutional, retrospective collection of pediatric brain tumor MRI scans, assembled for the 2023 edition of the BraTS 2023 Challenge with 99 patients. It can be seen as an out-of-distribution dataset for pretraining models (pediatric vs. brain), which also has T1, T1CE, T2, and FLAIR modalities. The original annotations are Non-Enhancing Tumor, Cystic Component, Edema, and Enhancing Tumor. For the final annotations for model training, the dataset has 3 labels: Nonenhancing Component (NC) (Label 1), Edema (ED) (Label 2),

and Enhancing Tumor (ET) (Label 3).

BraTS2023-MET. The BraTS2023-MET dataset, with 238 patients and T1, T1CE, T2, and FLAIR modalities, focuses on the segmentation of brain metastases on pre-treatment multiparametric MRI. The original annotations are: Necrosis, Edema, and Enhancing.

UPENN-GBM. The dataset includes MRI data from 630 patients diagnosed with de novo glioblastoma (adult, initial diagnosis). In this paper, we follow BrainMVP, adopting 127 patients. The original annotations are: Edema/Invasion, Enhancing Tumor, Necrosis.

ISLES22. ISLES (Ischemic Stroke Lesion Segmentation) is a recurring challenge for automatic segmentation of stroke lesions in MRI, with 238 patients. It has DWI, ADC, and FLAIR modalities. There is only one label (Stroke Lesion).

BraTS2018. BraTS2018 (Brain Tumor Segmentation Challenge 2018) dataset has expert manual annotations for sub-regions: Enhancing tumour (ET), Necrotic/non-enhancing tumour core (NCR/NET), Peritumoral edema (ED). These sub-regions can be grouped into ‘Whole Tumor’ (WT = ET + NCR/NET + ED), ‘Tumour Core’ (TC = ET + NCR/NET), and ‘Enhancing Tumour’ (ET) for evaluation.

ADNI. The ADNI is a large multicenter study launched in 2003 in the United States and Canada. In this paper, the dataset is classified as NC (Normal Control) and AD (Alzheimer’s disease), with 697 patients, and the T1 modality.

ADHD-200. ADHD-200 is a large multi-site open-access dataset compiled to support research on ADHD and related neuroimaging biomarkers. In this work, we adopt 767 patients with T1 modality. The dataset is classified as ADHD and NC.

The details of the pretraining and downstream datasets can be viewed in Table 2.

G.5. Details of Missing Modality

For the missing-modality task, we simplify the approach by using a random missing-modality training strategy. In this setup, 1 to 3 out of the 4 modalities will be randomly omitted. Our training framework utilizes the official BraTS21 implementation in Monai Swin UNETR. The hyperparameters are set as follows: learning rate= $1e-3$, batch size=1, epoch=300, optimizer=AdamW, and input volum size= $96 \times 96 \times 96$. During inference, a sliding window size of 0.75 and the same input volume size are used. A cosine annealing scheduler with warmup is also applied. The data split follows the official split in Monai Swin UNETR.

Tab. 11 shows the results of all missing-modality settings.

H. Pseudo Code of L_{intra} and L_{inter}

Algorithm 1 and 2 shows the pseudo code of L_{intra} and L_{inter} .

Algorithm 1 Intra-Instance Constraint $\mathcal{L}_{intra}$.

```

1: Input:
2:    $Z_i^h$  (output tokens from  $x_i^h$ )
3:    $Z_j^h$  (output tokens with shape  $K$  from  $x_j^h$ )
4:    $C_{i \rightarrow j}^h$  (Correspondence from modality  $i$  to  $j$ ,
5:     defaulting to an equivalent patch-index mapping.)
6:    $\omega$  (number of neighbors, default 5)
7:    $\delta$  (margin, default 0.1)
8:    $d(\cdot, \cdot)$  (distance metric, default ‘cosine’)
9: Calculate  $\mathcal{L}_{intra}$ :
10:   $\mathcal{L}_{intra} = 0$ 
11:  for  $h$  in  $H$ : {Each instance in the batch}
12:     $D_h \leftarrow d(Z_i^h, Z_j^h)$  {Compute pairwise distance between all tokens of  $Z_i^h$ }
13:     $\mathcal{P}_{i,h}^\omega \leftarrow D_h.arg\text{sort}()[1 : k + 1]$  {Find the  $Top_\omega$  nearest neighbors for each sample}
14:    for  $a$  in  $\text{arange}(K)$ :
15:       $a' \leftarrow C_{i \rightarrow j}^h(a)$ 
16:      for  $p$  in  $\mathcal{P}_{i,h}^\omega(a)$ :
17:         $p' \leftarrow C_{i \rightarrow j}^h(p)$ 
18:         $n \leftarrow \text{random\_select}(D_h.arg\text{sort}()[k + 2 :])$ 
19:         $n' \leftarrow C_{i \rightarrow j}^h(n)$ 
20:         $d_{pos} \leftarrow d(Z_j^h[a'], Z_j^h[p'])$  {Positive}
21:         $d_{neg} \leftarrow d(Z_j^h[a'], Z_j^h[n'])$  {Negative}
22:         $\mathcal{L}_{intra} \leftarrow \mathcal{L}_{intra} + \text{ReLU}(d_{pos} - d_{neg} + \delta)$ 
23:      end for
24:    end for
25:  end for
26:   $\mathcal{L}_{intra} \leftarrow \text{mean}(\mathcal{L}_{intra})$ 
27: return  $\mathcal{L}_{intra}$ 
    
```

Algorithm 2 Inter-instance Constraint $\mathcal{L}_{inter}$.

```

1: Input:
2:    $Z_i^h$  (output tokens from  $x_i^h$ )
3:    $Z_j^g$  (output tokens from  $x_j^g$ )
4:    $\omega$  (number of neighbors, default 5)
5:    $\delta$  (margin, default 0.1)
6:    $d(\cdot, \cdot)$  (distance metric, default 'cosine')
7: Calculate  $\mathcal{L}_{inter}$ :
8:   for  $(h, g)$  in  $H$ : {Paired instance in the batch}
9:      $D \leftarrow d(Z_i^h, Z_j^g)$  {Distance matrix between  $Z_i^h$  and  $Z_j^g$ }
10:     $D_{min1} \leftarrow D.\text{argmin}(\text{dim} = 1)$ 
11:     $D_{min0} \leftarrow D.\text{argmin}(\text{dim} = 0)$ 
12:     $\mathcal{M} \leftarrow \text{Arange}(K) == D_{min0}[D_{min1}]$ 
13:     $V_{i,j}^{h,g} \leftarrow \text{Arange}(K)[\mathcal{M}] \times D_{min1}[\mathcal{M}]$  {mutual nearest neighborhood}
14:     $D_{h,i} \leftarrow d(Z_i^h, Z_i^h)$ 
15:     $\mathcal{P}_{i \rightarrow j, h \rightarrow g}^\omega \leftarrow D_{h,i}.\text{argsort}()[1 : k + 1]$  {Find the  $Top_\omega$  nearest neighbors for each sample}
16:    for  $(a, a')$  in  $V_{i,j}^{h,g}$ :
17:      for  $p$  in  $\mathcal{P}_{i \rightarrow j, h \rightarrow g}^\omega(a)$ :
18:         $p' \leftarrow \text{find}(p, p')$  in  $V_{i,j}^{h,g}$ 
19:         $n \leftarrow \text{random\_select}(D_{h,i}.\text{argsort}()[1 : k + 2 :])$ 
20:         $n' \leftarrow \text{find}(n, n')$  in  $V_{i,j}^{h,g}$ 
21:         $d_{pos} \leftarrow d(Z_j^g[a'], Z_j^g[p'])$  {Positive}
22:         $d_{neg} \leftarrow d(Z_j^g[a'], Z_j^g[n'])$  {Negative}
23:         $\mathcal{L}_{inter} \leftarrow \mathcal{L}_{inter} + \text{ReLU}(d_{pos} - d_{neg} + \delta)$ 
24:      end for
25:    end for
26:  end for
27:   $\mathcal{L}_{inter} \leftarrow \text{mean}(\mathcal{L}_{inter})$ 
28: return  $\mathcal{L}_{intra}$ 
    
```
